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Techniques to derive and clean acceleration and deceleration data of athlete tracking technologies in team sports: A scoping review

Susanne Ellens[a,b], Kane Middleton[a,b], Paul B. Gastin[a,b] and Matthew C. Varley[a,b]

[a]Sport and Exercise Science, School of Allied Health, Human Services & Sport, La Trobe University, Melbourne, VIC, Australia; [b]La Trobe Sport and Exercise Medicine Research Centre, La Trobe University, Melbourne, VIC, Australia

ABSTRACT

The application of acceleration and deceleration data as a measure of an athlete's physical performance is common practice in team sports. Acceleration and deceleration are monitored with athlete tracking technologies during training and games to quantify training load, prevent injury and enhance performance. However, inconsistencies exist throughout the literature in the reported methodological procedures used to quantify acceleration and deceleration. The object of this review was to systematically map and provide a summary of the methodological procedures being used on acceleration and deceleration data obtained from athlete tracking technologies in team sports and describe the applications of the data. Systematic searches of multiple databases were undertaken. To be included, studies must have investigated full body acceleration and/or deceleration data of athlete tracking technologies. The search identified 276 eligible studies. Most studies (60%) did not provide information on how the data was derived and what sequence of steps were taken to clean the data. Acceleration and deceleration data were commonly applied to quantify and describe movement demands using effort metrics. This scoping review identified research gaps in the methodological procedures and deriving and cleaning techniques that warrant future research focussing on their effect on acceleration and deceleration data.

ARTICLE HISTORY
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KEYWORDS
Acceleration; deceleration; tracking system; data filter; GPS; team sport

1 Introduction

The application of acceleration data as a measure of an athlete's physical performance is common practice in team sports (Bradley et al., 2010; Cunningham, 2016; Fox et al., 2020; Gabbett, 2010; Kelly et al., 2019). Acceleration can be defined as the rate of change of velocity over time, where positive acceleration is an increase in velocity whilst negative acceleration is a decrease in velocity (better known and referred to in this paper as deceleration). Researchers and practitioners are interested in quantifying high rates of acceleration and deceleration (e.g., $> \pm 2\ m \cdot s^{-2}$, $> \pm 3\ m \cdot s^{-2}$, etc) as they are considered more energetically demanding than when they are low (Osgnach et al., 2009) and may have a critical impact on a player's performance. High rates of acceleration and deceleration, for example, allow a player to reach the ball before an opponent or stop in time to avoid a tackle. In team sports, athletes frequently accelerate and decelerate at a high rate as the nature of the game requires many changes of a player's velocity over a short duration (Harper et al., 2019a). Australian football players, for example, perform on average 82 high acceleration efforts ($\geq 2.78\ m \cdot s^{-2}$) in a single match (Varley et al., 2014). Therefore, team sports researchers and practitioners monitor acceleration and deceleration during training and games for quantifying training load, injury prevention and performance enhancement (Akenhead & Nassis, 2016; Harper et al., 2019b) as they occur frequently, are energetically demanding and are considered important to assess athletes' performance.

Acceleration and deceleration data are used in a variety of studies, such as descriptive and validation studies (Linke et al., 2018; Ryan et al., 2020). The data in these studies are often used to describe the movements of players, player positions and teams, often with the inclusion of contextual variables as they influence the data (Barrera et al., 2021; Lago, 2009; Sullivan et al., 2014). Examples of contextual variables include actions that precede or follow acceleration and deceleration, opponent level and match score-line (Barrera et al., 2021; Lago, 2009; Sullivan et al., 2014). The data can be applied in a variety of contexts, such as to mimic specific movement demands or quantify training load (Akenhead et al., 2016; Suarez-Arrones et al., 2016; Wisbey et al., 2010), where variables related to acceleration and deceleration activities are one of the most commonly used variables to monitor training load (Akenhead & Nassis, 2016). Furthermore, acceleration and deceleration data can be used to enhance player fitness or assess performance (Buchheit et al., 2010; Malone & Collins, 2017), all together resulting in a broad applied setting of the data.

Acceleration and deceleration are monitored in team sports with athlete tracking technologies, in particular, Global Positioning Systems (GPS), radio-based local positioning systems (LPS), optical positioning systems and inertial measurement units (consisting of an accelerometer, gyroscope and magnetometer; Buchheit, Allen et al., 2014; Christopher et al., 2008). To calculate acceleration and deceleration, athlete tracking technologies use different deriving techniques specific to each manufacturer's software. Data deriving techniques are

CONTACT Susanne Ellens s.ellens@latrobe.edu.au Sport and Exercise Science, School of Allied Health, Human Services & Sport, La Trobe University, Melbourne, VIC, Australia

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defined as a sequence of steps, logical or computational, that lead from one result to another. There are three different deriving techniques used to calculate acceleration and deceleration data from athlete tracking technologies: 1) double differentiation of position data, 2) differentiation of Doppler shift velocity data (velocity measured by the rate of change in satellites' radio frequency signals; Schutz & Herren, 2000), 3) direct measurement from an accelerometer. After the acceleration and deceleration data has been derived, different data cleaning techniques are typically applied in the manufacturer's software to clean the noise in the raw data. Examples of data cleaning techniques include data filtering (e.g., Butterworth, Bandpass, Wavelet transform), data down sampling, data up sampling and gap filling (e.g., interpolation). Different data processing procedures (deriving and cleaning) can directly change the acceleration and deceleration values (Winter, 2009). This would lead to different outputs if the same data was analysed using different methods.

After acceleration and deceleration data have been derived and cleaned, various acceleration and deceleration metrics have been established through custom analysis by researchers or within the manufacturer's software. These metrics are the final acceleration/deceleration output often reported and help practitioners to interpret the data. Examples include, the total or average number of acceleration and/or deceleration efforts performed (Akiyama et al., 2019; Oliva-Lozano, 2020; Ryan et al., 2020), the distance covered while accelerating or decelerating (Oliva-Lozano, 2020; Ryan et al., 2020) and the duration an athlete spends accelerating (Ryan et al., 2020). Different athlete tracking technologies, models and software versions can have different data processing procedures (deriving and cleaning) and metric options (such as choosing custom thresholds and identifying a minimum effort duration), which directly change the final acceleration and deceleration metrics (Buchheit, Haddad et al., 2014; Varley et al., 2017). Furthermore, acceleration and deceleration movements in team sports vary in duration, magnitude and the velocity at which they commence and end (Varley & Aughey, 2013) as team sports consist of chaotic, unpredictable and continuously changing movements (Polglaze, 2018). Therefore, large variations exist in acceleration and deceleration metrics reported across the literature where different types, brands, models and software versions of tracking technologies have been used and different processing procedures applied (Buchheit, Haddad et al., 2014; Varley et al., 2017). For example, differences in the reported number of high acceleration efforts during a 90 minute soccer match have been as large as 269 efforts between software versions (Varley et al., 2017), 35 efforts between tracking technologies (Buchheit, Allen et al., 2014) and 291 efforts when different metric options have been used (Varley et al., 2017). As such, the acceleration and deceleration metrics are variable and research mapping the specific details of the used tracking technologies and metrics is wanted.

There is a substantial amount of literature profiling acceleration and deceleration in team sports (Bradley et al., 2010; Cunningham, 2016; Fox et al., 2020; Gabbett, 2010; Kelly et al., 2019). However, inconsistencies exist throughout the literature in the reported methodological procedures (i.e., specific details of tracking technologies, deriving and cleaning techniques and metric options) used to quantify acceleration and deceleration. Furthermore, a rationale is often not provided to justify why methodological procedures were selected. Given the variation, importance and frequent use of acceleration and deceleration metrics in team sports, there is a need to systematically map and provide an overview of the methodological procedures being used.

The aim of this scoping review was to identify and characterise methodological procedures being used on acceleration and deceleration data obtained from athlete tracking technologies in team sports. A secondary aim was to describe the applications of acceleration and deceleration data of athlete tracking technologies in team sports. This review will inform practitioners and researchers of the used methodological procedures for acceleration and deceleration data of athlete tracking technologies in team sports and highlight the potential gaps, inconsistencies and improvements within the literature.

2 Methods

The protocol for the scoping review was drafted and revised by the research team using the reporting items of the PROSPERO registration form. The final protocol was registered with Open Science Framework (osf.io/6anpg).

2.1 Search strategy

A systematic search of the literature was performed based on the PRISMA-ScR guidelines for scoping reviews (Tricco et al., 2018) and JBI scoping review methodology (Peters et al., 2020). The databases of PubMed, SPORTDiscus, Scopus and Web of Science were used for the literature search conducted by one author from the time of database inception to August 2020. Databases were searched based on articles' title, abstract and keywords. The search strategy only included terms relating to or describing the review aims, which resulted in the following search string:

(sport OR match* OR game* OR training* OR team-sport* OR soccer OR football OR futsal OR rugby OR AFL OR NRL OR hurling OR basketball OR netball OR handball OR hockey OR lacrosse OR korfball OR floorball OR volleyball OR cricket OR baseball OR softball) AND (accelera* OR decelera*) AND ("tracking device*" OR "tracking system*" OR "positioning system*" OR "optical system*" OR "GPS" OR "LPS" OR "microtechnolog*" OR "micromechanical-electrical system" OR microsensor* OR "match analysis system*" OR "multi-camera system*" OR acceleromet* OR "inertial measurement unit")*

Publications needed to be peer reviewed, no further publication type restrictions for the search were in place, as this was a scoping review, all types of publications were relevant including published articles, conference proceedings, reports and textbook chapters.

2.2 Source of evidence selection

The review was conducted in six stages, where stages 3–6 have been performed with the Covidence systematic review software (Veritas Health Innovation, Melbourne, Australia): (1) initial

Table 1. Study inclusion and exclusion criteria.

Inclusion criteria	Exclusion criteria
• Original peer reviewed published manuscripts • Field-based or court-based team sports • Team sport athletes of all ages and all levels • Investigated athletes' full body acceleration and/or deceleration data of athlete tracking technologies	• Athletes with physical or mental disability • Water-based team sports • No use of athlete tracking technologies • No acceleration or deceleration measures • Segmental acceleration or deceleration • Studies using the same data set from an earlier publication • Cannot access full text in English

search of two databases with title screening to identify other keywords (conducted by SE); (2) second search of all databases (conducted by SE); (3) removal of duplicate papers (conducted by SE); (4) title screening (conducted by SE); (5) abstract screening (conducted by SE & MV) to assess for eligibility; (6) full-text examination for assurance that inclusion and exclusion criteria (Table 1) were met (conducted by SE & MV). Any disagreements were solved by consensus or by the decision of a third reviewer. The remaining articles were included in the scoping review.

A custom-made extraction template was created in Covidence systematic review software by two researchers. Data collection and extraction of the relevant information from the included full-text articles were conducted by two researchers. The extracted data were categorised into sample studied (gender, age, sport, level), study design (split in descriptive and analytical studies) athlete tracking technologies (device brand, model details, sampling frequency, software version), data output (e.g., efforts, distance), data processing (deriving technique, cleaning technique, justification and rationale for processing) and application of the acceleration and deceleration data.

3 Results

After duplicate removal, a total of 5015 articles were identified from the literature search using the databases and reference lists. Based on title and abstract screening, 4661 articles were excluded, with 354 full-text articles remaining to be assessed for eligibility. A total of 276 articles met the criteria to be included in this scoping review (Figure 1).

3.1 Study characteristics

The number of articles published per year from 2010 steadily increased until 2017 and more than doubled in 2018 with 165 articles published between 2018 and 2020 (Figure 2). The majority of the studies used acceleration measures (n = 270) compared to deceleration measures (n = 148), whereof 142 used both measures. The studies included were categorised as descriptive studies (n = 264) and analytical studies (n = 12).

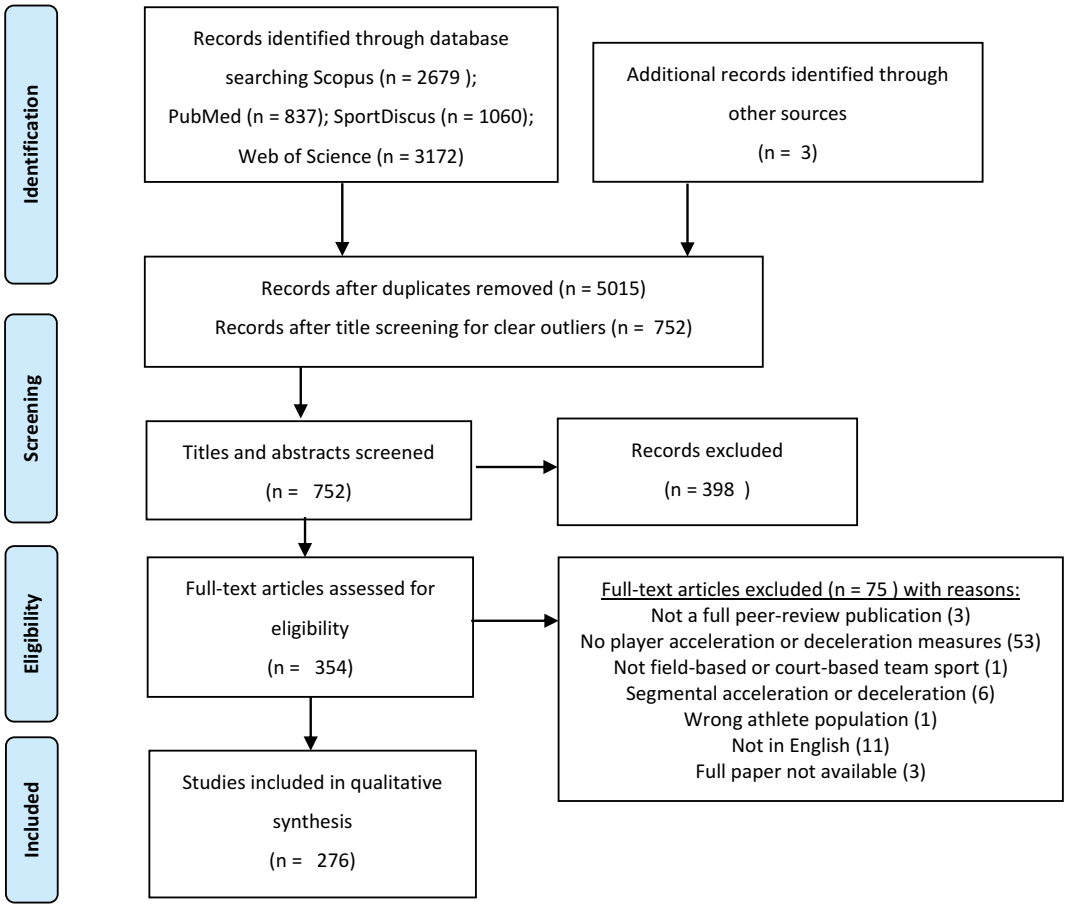

Figure 1. PRISMA flow diagram of the study selection process.

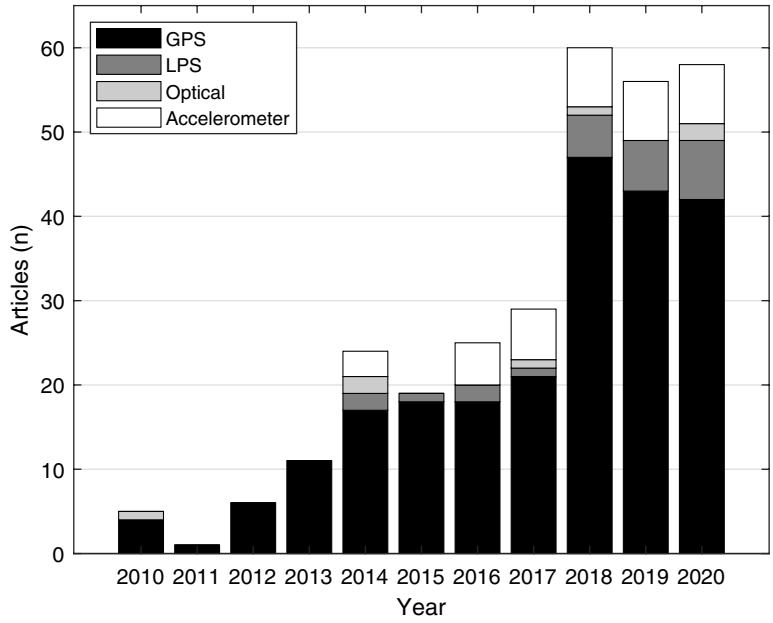

Figure 2. Number of articles published per year of articles included in the review, subcategorized by used tracking technology. GPS = global positioning system device, lps = local positioning system, optical = optical positioning system.

Research was conducted in 20 different sports, with the majority being soccer (n = 144) and four studies used simulated team sport protocols (Figure 3).

3.2 Tracking technologies

Of the athlete tracking technologies used to measure acceleration and deceleration data 79% of the included studies used GPS devices (n = 218) followed by accelerometers (n = 23), LPS (n = 22) and Optical positioning systems (n = 7). Twelve studies specified more than one tracking technology (GPS or LPS and an accelerometer) but did not indicate which tracking technology had been used to collect the reported acceleration and/or deceleration data. A total of 21 different athlete tracking technology brands were used. Almost half of the included GPS studies used the athlete tracking technology brand Catapult Sports (45%), while 24% used GPSports and 13% used STATSports. The remaining 18% used K-Sport (5%), VX Sport (4%), Polar Electro (2.5%), Realtrack systems (2%), Digital simulation (2%), JOHAN Sports (1%), Exelio SRL (1%) or Advanced Sport Instrument Sarl (0.5%). The majority (65%) of the studies using an accelerometer used the brand Catapult Sports, followed by GPSports (9%), StatSport (4%), Polar Electro (4%), ActiGraph (4%), Noraxon (4%), Delsys (4%) or Analog Devices (4%). The studies using a LPS used the

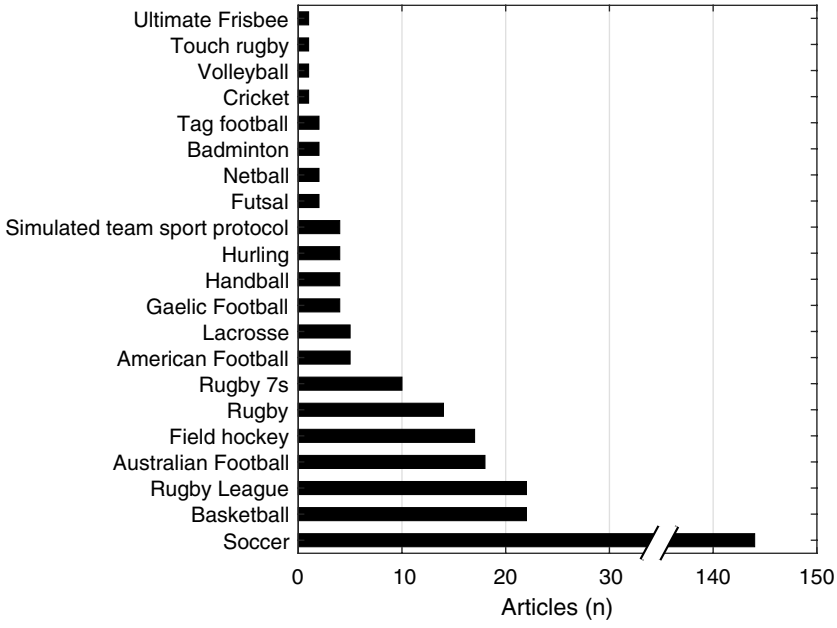

Figure 3. Sports of included articles.

brands Realtrack systems (32%), Chyron Hego (18%), ZXY Sport Tracking System (18%), Inmotio (18%), Catapult Sports (9%) or Wireless Ad Hoc System for Positioning (5%). The studies using an Optical tracking technology used the brands ChyronHego (29%), Prozone (29%), STATS SportVU (14%) or Sport Universal Process (14%). The majority of the included studies sampled the data at 10 Hz (n = 141), with the remaining studies sampling either between 15 and 45 Hz (n = 62), <10 Hz (n = 48) or ≥100 Hz (n = 33). Five of the studies that sampled the data at 5 Hz interpolated the data to 15 Hz. Relevant data relating to the athlete tracking technologies extracted from the articles are presented in Table 2.

3.3 Data processing

Of all studies, 66% did not mention how the acceleration and/or deceleration data were derived. The majority of the studies derived the data from velocity (n = 50). Of these, 14 specified that this was Doppler shift velocity, whereas the other 36 did not specify how velocity was calculated. See, Figure 4 for an overview of the used deriving techniques.

Most of the articles (88%) did not mention any data cleaning technique. Of the articles which reported their data cleaning technique (n = 32), 9 articles specified that the manufacturer's software applied a filter to the data, but no specific filter was mentioned. Four articles mentioned the use of a filter to clean the data but did not provide enough details for their filtering process to be replicated. An overview of the used cleaning techniques is presented in Figure 5. Most studies (n = 26) did not report any rationale for using the data cleaning technique or referred to a study using the data cleaning method which also had no rationale for using it. Studies that did provide a rationale, either stated that it was the same technique used by the manufacturer's software (n = 5),

Table 2. Tracking technology details of the included studies.

Tracking system	Device brand	Model details	Sampling frequency	Software	Studies	Number of studies
GPS (n=218)	Catapult Sports (n=99)	OptimEye S5	10	Sprint; OpenField	[1-15]	n=15
		OptimEye S5B	10	Sprint	[16, 17]	n=2
		OptimEye X4	10	OpenField	[18-20]	n=3
		MinimaxX	5, 10	Logan Plus; Sprint; Team AMS	[21-33]	n=13
		MinimaxX S3	5	Sprint	[34, 35]	n=2
		MinimaxX S4	10	Sprint; Openfield	[5, 10, 34-48]	n=17
		MinimaxX X4	10	Open Field	[49]	n=1
		MinimaxX V4	10	Logan Plus; Sprint; Team AMS	[18, 50-59]	n=11
		MinimaxX S5	10	Sprint	[60]	n=1
		MinimaxX Team Sports 2.0	5	Logan Plus	[61-63]	n=3
		MinimaxX Team Sports 2.5	5	Logan Plus; Sprint	[64-66]	n=3
		MinimaxX Team S5	10	Openfield	[67]	n=1
		Team 6.5	10	Logan Plus	[68]	n=1
		PlayerTek	10	PlayerTek Cloud	[69]	n=1
		X	**X**	**X**	[70-99]	n=30
	GPSports (n=53)	SPI 10	1	Sports Tracker Analysis	[100]	n=1
		SPI-Elite	1, 10	Team AMS; Sports Tracker Analysis	[100-106]	n=7
		SPI Pro	5, 10	Team AMS	[107-109]	n=3
		SPI Pro X	5, 10, 15	Team AMS	[110-123]	n=14
		SPI Pro X2	5	Team AMS	[124]	n=1
		SPI-pro W2b	15	Team AMS	[125]	n=1
		SPI-HPU	5, 15	Team AMS	[126-132]	n=7
		EVO	10	GPSports Console	[133]	n=1
		X	**X**	**X**	[96, 134-152]	n=20
	StatSports (n=29)	Viper Pod	10	Viper PSA software	[153-167]	n=15
		Viper Pod 2	10	Viper PSA software	[168-173]	n=6
		Apex	10, 18	Apex Software	[174-176]	n=3
		X	**X**	**X**	[177-181]	n=5
	RealTrack Systems (n=5)	WIMU	5	Qüiko	[182]	n=1
		WIMU Pro	10	SPRO	[183-186]	n=4
	K-Sport (n=10)	K-GPS	10	K-Fitness	[187-189]	n=3
		X	**X**	**X**	[190-196]	n=7
	VX Sport(n=8)	VX330a	4	VX Sport View	[197, 198]	n=2
		X	**X**	**X**	[199-204]	n=6
	Polar Electro(n=5)	Polar Team Pro	10	Polar Team Pro web service	[205]	n=1
		Polar Team Pro	**X**	**X**	[206-209]	n=4
GPS (n=218)	Digital simulation(n=4)	SensorEverywhere	8, 16	Sensoreverywhere Analyser	[210, 211]	n=2
		SensorEverywhere V2	16	Sensoreverywhere Analyser	[212, 213]	n=2
	JOHAN Sports(n=2)	**X**	10	JOHAN Sports web application, JOHAN application	[214, 215]	n=2
	Exelio SRL(n=2)	GPEXE GK	18.18	**X**	[216]	n=1
		GPEXE System	18.18	**X**	[217]	n=1
	Advanced Sport Instrument Sarl (n=1)	FieldWiz	10	**X**	[218]	n=1

(*Continued*)

Table 2. (Continued).

Tracking system	Device brand	Model details	Sampling frequency	Software	Studies	Number of studies
LPS (n=22)	Catapult Sports(n=2)	Clearsky T6	10	Openfield	[219, 220]	n=2
	RealTrack Systems(n=7)	WIMU PRO	18, 20	SPRO	[221-223]	n=3
		WIMU PRO	**X**	**X**	[224-227]	n=4
	ChyronHego(n=4)	ZXY Tracking system	20	ZXY Sport Tracking	[228]	n=1
		ZXY Tracking system	20	**X**	[229-231]	n=3
	ZXY SportTracking AS(n=4)	ZXY Sport Chip	20, 40	**X**	[232, 233]	n=2
		X	**X**	**X**	[234, 235]	n=2
	Inmotio(n=4)	**X**	45, 45.45	Inmotio Object tracking, Inmotio Client	[111, 124]	n=2
		X	28, 45	**X**	[236, 237]	n=2
	Wireless Ad Hoc System for Positioning (n=1)	**X**	10	WheresBruce	[238]	n=1
Optical (n=7)	ChyronHego(n=2)	TRACAB	**X**	**X**	[173, 239]	n=2
	Prozone(n=2)	Prozone Version 3.0	**X**	Stadium Manager	[240]	n=1
		X	10	**X**	[124]	n=1
	STATS SportVU(n=1)	**X**	16	STATS SportVU version 2.12.0	[111]	n=1
	Sport Universal Process (n=1)	Amisco Pro	10	Amisco Viewer	[241]	n=1
	X (n=1)	**X**	25	Optical Player Tracking, Australian Institute of Sport	[242]	n=1
Accelerometer (n=23)	Catapult Sports(n=15)	S5	100	Sprint, OpenField	[1, 243-245]	n=4
		MinimaxX S4	100	Sprint	[246-248]	n=3
		Clearsky T6	100	OpenField	[249]	n=1
		X	**X**	**X**	[250-256]	n=7
Accelerometer (n=23)	GPSports(n=2)	SPI Pro X	100	Team AMS	[257]	n=1
		SPI HPU	100	**X**	[258]	n=1
	StatSports (n=1)	Viper	100	**X**	[259]	n=1
	Polar Elctro (n=1)	Polar Team Pro	200	**X**	[260]	n=1
	ActiGraph (n=1)	GTX3	100	**X**	[261]	n=1
	Noraxon (n=1)	DTS 3D	500	**X**	[262]	n=1
	Delsys(n=1)	Delsys Trigno Wireless EMG System	148	**X**	[263]	n=1
	Analog Devices (n=1)	ADXL326	100	Viper	[264]	n=1
Unclear (n=12)	Catapult Sports(n=2)	MinimaxX Team Sport 2.5	10, 100	Sprint	[265]	n=1
		MinimaxX	5, 100	Sprint	[266]	n=1
	GPSports (n=4)	SPI Pro X	5, 100	Team AMS	[267-270]	n=4
	StatSports (n=1)	Viper	10, 100	Viper PSA software	[271]	n=1
	RealTrack Systems(n=2)	WIMU pro	18, 100	SPRO	[272, 273]	n=2
	VX Sport(n=2)	VX235	10, 100	**X**	[274]	n=1
	Polar Electro(n=2)	Polar team Pro	10, 100, 200	**X**	[275, 276]	n=2

Note: X = Information of model details, sampling frequency and/or Software version not provided; Tracking system category Unclear indicates that more than one tracking technology had been specified and it had not been indicated which tracking technology had been used to collect acceleration and/or deceleration data.

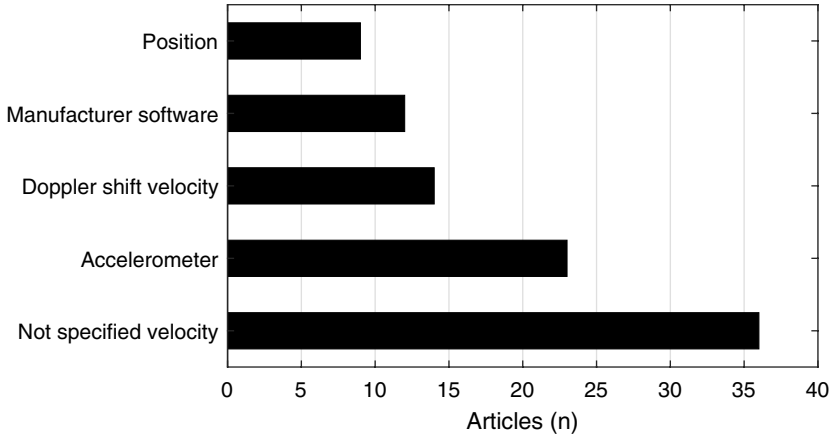

Figure 4. Techniques used to derive acceleration and/or deceleration data of tracking technologies of included articles.

referred to a specific method of a study (n = 3), stated that it was a common filtering technique in biomechanics (n = 1) or that it was commonly applied in human motion (n = 1). Data deriving and cleaning techniques categorised per metric extracted from the studies are presented in Table 3.

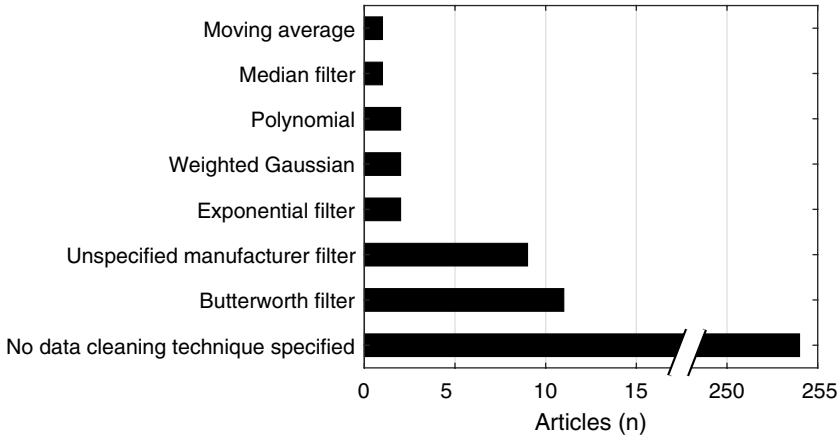

Figure 5. Techniques used to clean acceleration and/or deceleration data of tracking technologies of included articles.

3.4 Metrics

It has been identified that researchers and practitioners report acceleration and deceleration in four different metrics/ways: 1) Distance covered while accelerating and/or decelerating; 2) Number of acceleration or deceleration efforts performed; 3) Time spent accelerating and/or decelerating; and 4) Maximum, minimum, average or total acceleration and/or deceleration (Figure 6). Some of the metrics were defined by different set criteria, which the acceleration and/or deceleration data need to meet to be included in the metrics. For example, distance covered at high rates of acceleration (>3 $m \cdot s^{-2}$). The most used acceleration and/or deceleration metric was number of efforts performed. Of the studies using number of efforts performed, more than half (59%) only specified the effort by a set threshold and 30% also used a minimum duration (Table 4). Studies reporting high-intensity acceleration and deceleration efforts (>2.78 $m \cdot s^{-2}$ or < −2.78 $m \cdot s^{-2}$) per player per match (not position specific and only for players playing full time) were variable and inconsistent within and between sports (Figure 7). Further details of reported acceleration and deceleration efforts across the literature can be found elsewhere (Harper et al., 2019b).

3.5 Applications

Researchers have applied the acceleration and deceleration data and metrics in four different ways: 1) investigating the validity and/or reliability of acceleration and deceleration metrics; 2) quantifying and describing acceleration and deceleration data to determine the movement profile of athletes; 3) quantifying acceleration and deceleration data to measure external load of athletes; 4) determine the influence of acceleration and deceleration on the recovery and injury risk of athletes. Most of the studies (57%) applied the acceleration and deceleration data to quantify and describe movement demands (Figure 8).

4 Discussion

This scoping review summarised current research and identified gaps, inconsistencies and improvements related to the methodological procedures being used on acceleration and deceleration data obtained from athlete tracking technologies in team sports. Overall, the search yielded 276 studies, whereof the majority measured acceleration and deceleration with GPS devices, derived the data from velocity and filtered it with a low-pass Butterworth filter. A research gap was identified in the data processing procedures, specifically, on how acceleration and deceleration data is derived from athlete tracking technologies and what sequence of steps have been taken to clean the data. Further, it has been identified that method sections are lacking specific information regarding the data processing procedures and used athlete tracking technologies, which prevents replicability of the studies and comparison of findings.

Studies investigating acceleration and deceleration data increased exponentially within the last few years, where the most researched sport was soccer (n = 144), leading by 122 articles from the second most researched sports basketball and rugby league. Soccer being the most popular sport in the world could explain the higher amount of papers in this sport. Improvements in athlete tracking technologies, together with approval to allow tracking technologies within official soccer match play in 2015, may explain the sudden exponential growth of published research in this area from 2018.

Research that has investigated acceleration is much higher (47% more) compared to deceleration. This could be explained by the poor validity and reliability of the deceleration measure (Buchheit, Haddad et al., 2014; M. Varley et al., 2012). The accuracy of GPS devices in measuring deceleration has been reported to have overestimations of up to 19.3% (M. Varley et al., 2012) and between-unit variations of up to 56% (Buchheit, Haddad et al., 2014). However, deceleration is an important movement to consider as it is a key element of team sports which require

Table 3. Data deriving and cleaning techniques of the included studies categorised per metric used.

Metric	Tracking system	Acceleration/deceleration data derived from:	Acceleration/deceleration data cleaned with:	Rationale	Studies
Acceleration and/ or deceleration efforts	GPS	Position	3 Hz Butterworth low pass filter	Referred to other study [1]	[2]
		Velocity	Butterworth low-pass second-order (cutoff frequency: 1 Hz) double phase-lag filter	**X**	[3]
			The interval over which acceleration was calculated from velocity in the manufacturer software (smoothing filter width) was set at 0.2 seconds.	Referred to other study [4]	[5]
			Use of cleaning mentioned but no specific details given	**X**	[6-8]
			X	**X**	[9-26]
		Doppler	0.3-second smoothing filter	**X**	[27]
			Sprint software: Raw velocity data filtered using a Median filter then acceleration derived; Openfield software: Raw velocity data filtered using an exponential filter, then acceleration derived and filtered further using an exponential filter.	As per manufacturer software of tracking system	[4]
			X	**X**	[28-35]
		X	Fourth-order low-pass Butterworth filter with a 1Hz cut-off frequency was applied to each players velocity-time curve	Referred to other study [36]	[37]
			Smoothing filter of 0.2s	**X**	[38]
			15 Hz interpolation	**X**	[39]
			Exponential filter	**X**	[40]
			15 Hz interpolation	**X**	[41]
			Use of cleaning mentioned but no specific details given	**X**	[42]
			X	**X**	[43-173]
	LPS	Position	**X**	**X**	[174-176]
		X	Weighted Gaussian average filter set at 85%	As recommended by the manufacturer of tracking system	[43]
			Butterworth low-pass filter (order 5) with a frequency cut-off of 0.02	**X**	[177]
			X	**X**	[178-190]
	Optical	Position	Butterworth low-pass second-order (cut-off frequency: 1 Hz) double phase-lag filter was used after each derivation.	**X**	[191]
Acceleration and/ or deceleration efforts	Optical	Position	Piece wise cubic polynomial using the midpoint for each one-second epoch	Provides a compact representation of the large-scale tracking data, and also offers a smoothing function for noisy data	[192]
		X	**X**	**X**	[43, 56, 193]
	Accelerometer	N.A.	Polynomial least squares fit	As per manufacturer software of tracking system	[194]
			15 Hz interpolation	**X**	[41]
			X	**X**	[76, 90, 99, 137, 150, 152, 180, 184, 196-210]

(*Continued*)

Table 3. (Continued).

Metric	Tracking system	Acceleration/deceleration data derived from:	Acceleration/deceleration data cleaned with:	Rationale	Studies
Distance run accelerating and/or decelerating	GPS	Position	X	X	[211]
		Doppler	15 Hz interpolation	X	[212]
			X	X	[213]
		Velocity	Use of cleaning mentioned but no specific details given	X	[6]
			X	X	[9-12, 25, 26, 214]
		X	Smoothing filter of 0.5 s was applied to GPS data using the manufacturer's software	X	[215]
			15 Hz interpolation	X	[216]
			X	X	[60, 72, 73, 85, 102, 106, 108, 114, 124, 126, 133, 136, 147, 149, 170-172, 217-243]
	LPS	X	Weighted Gaussian average filter set at 85%	As recommended by the manufacturer of tracking system	[212]
		X	X	X	[244]
	Optical	Position	Piece wise cubic polynomial using the midpoint for each one-second epoch	Provides a compact representation of the large-scale tracking data, and also offers a smoothing function for noisy data	[192]
		X	X	X	[212]
	Accelerometer	N.A.	15 Hz interpolation	X	[216]
			X	X	[197, 221, 245]
Time spent accelerating and/or decelerating	GPS	Velocity	Use of cleaning mentioned but no specific details given	X	[6, 246]
			X	X	[10]
		X	X	X	[73, 106, 108, 247]
	Optical	Position	Butterworth low-pass second-order (cut-off frequency: 1 Hz) double phase-lag filter was used after each derivation.	X	[191]
	Accelerometer	N.A.	Butterworth low-pass filter (zero-lag, 4th order, 15 Hz cut-off frequency)	Referred to other study [248]	[249]

(*Continued*)

Table 3. (Continued).

Metric	Tracking system	Acceleration/deceleration data derived from:	Acceleration/deceleration data cleaned with:	Rationale	Studies
Max/min/average acceleration and/or deceleration	GPS	Velocity	interpolated to 15 Hz	Part of the manufacturer software	[250]
			The interval over which acceleration was calculated from velocity in the manufacturer software (smoothing filter width) was set at 0.2 seconds.	Referred to other study [4]	[5]
			X	X	[26, 251-259]
		Doppler	0.5-s moving average	As per manufacturer software of tracking system	[260]
			15 Hz interpolation	X	[212]
			X	X	[261]
		X	Velocity-time curves were linearly interpolated to 15 Hz, and a fourth-order Butterworth filter applied with a 1-Hz cut-off frequency	X	[262]
			15 Hz interpolation	X	[263]
			Use of cleaning mentioned but no specific details given	X	[264]
			X	X	[43, 58, 72, 73, 111, 148, 173, 232, 265, 266]
	LPS	Position	X	X	[267]
		X	Weighted Gaussian average filter set at 85%	As recommended by the manufacturer of tracking system	[43, 212]
			X	X	[183, 186-189, 268-271]
	Optical	Position	Butterworth low-pass second-order (cut-off frequency: 1 Hz) double phase-lag filter was used after each derivation.	X	[191]
			Raw position data was filtered using a 4th order low-pass Butterworth filter with 1HZ cut-off frequency	Referred to other study [272]	[273]
		X	X	X	[43, 193]
Max/min/average acceleration and/or deceleration	Accelerometer	N.A.	15 Hz interpolation	X	[216]
			Fourth order low pass Butterworth filter with a cut-off frequency of 9.97 Hz.	The cut-off frequency was chosen after residual analysis (SD<1.0 Hz) and on the basis of the lowest mean bias compared to the Motion Analysis system [272].	[195]
			Fourth order low pass Butterworth filter with a cut-off frequency of 6.23 Hz;	The cut-off frequency was chosen after residual analysis (SD<1.0 Hz) and on the basis of the lowest mean bias compared to the Motion Analysis system [272]	[274]
			X	The manufacturer was contacted and denied the request to provide the GPSports device acceleration filtering methods. Therefore, this data was not filtered after it was exported from the devices software	[275]
				X	[208]

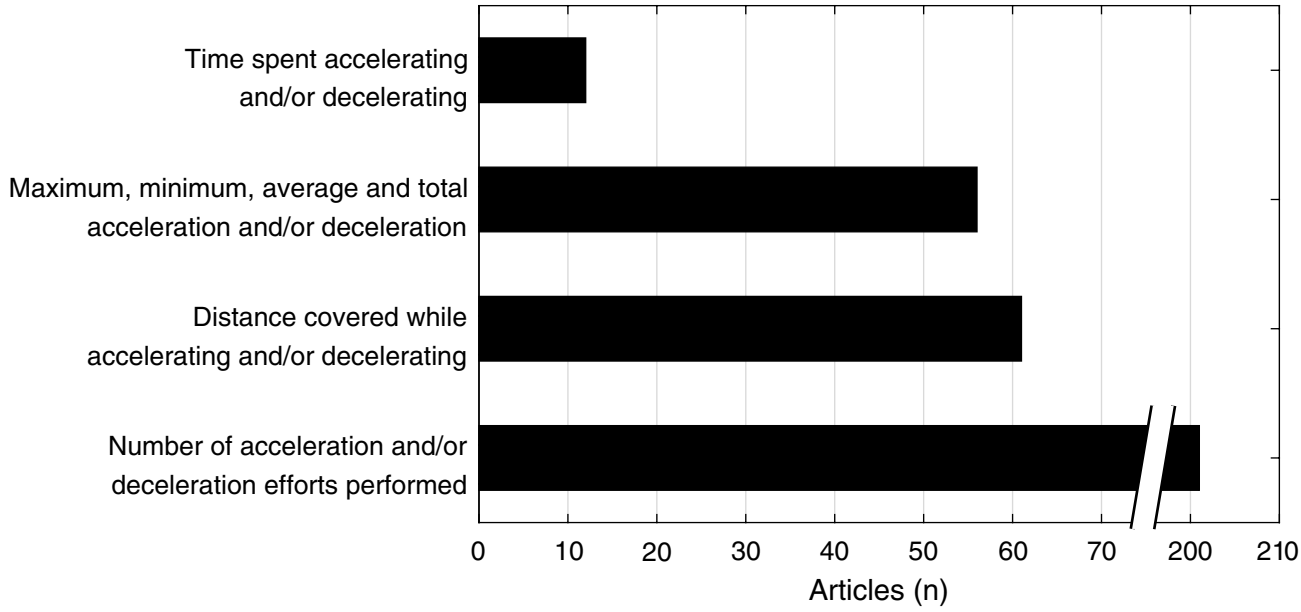

Figure 6. Different metrics/ways acceleration and deceleration have been reported of included articles.

Table 4. Different criteria used within the literature to define acceleration and deceleration efforts. ○ = Examples of the effort criteria.

Effort criteria	Number of studies using the effort criteria
Effort only requires acceleration to be above a set threshold	
o E.g. Acceleration >3 m·s^{-2}	119
Effort requires acceleration to remain above a set threshold for a minimum duration	
o E.g. Acceleration >3 m·s^{-2} for 0.5 seconds	60
Effort requires acceleration to remain above a set threshold for a minimum duration and ends when the acceleration reaches zero.	
o E.g. Acceleration >3 m·s^{-2} for 0.5 seconds until acceleration = 0 m·s^{-2}	10
Unspecified	12

repeated changes of a player's velocity over a short duration (Harper et al., 2019a). Furthermore, higher deceleration distance of team sport athletes has been associated with an elevated creatine kinase concentration, which is an indication of muscle damage (Young et al., 2012). As such, deceleration is an important measure in team sports that is commonly overlooked. Therefore, future research should investigate ways to improve the validity and reliability of deceleration data.

Global Positioning System devices were the most used athlete tracking technology to measure acceleration and deceleration data. The GPS devices do not require a labour-intensive setup and are not restricted to a location (whilst LPS and Optical tracking technologies are, as they need to be installed

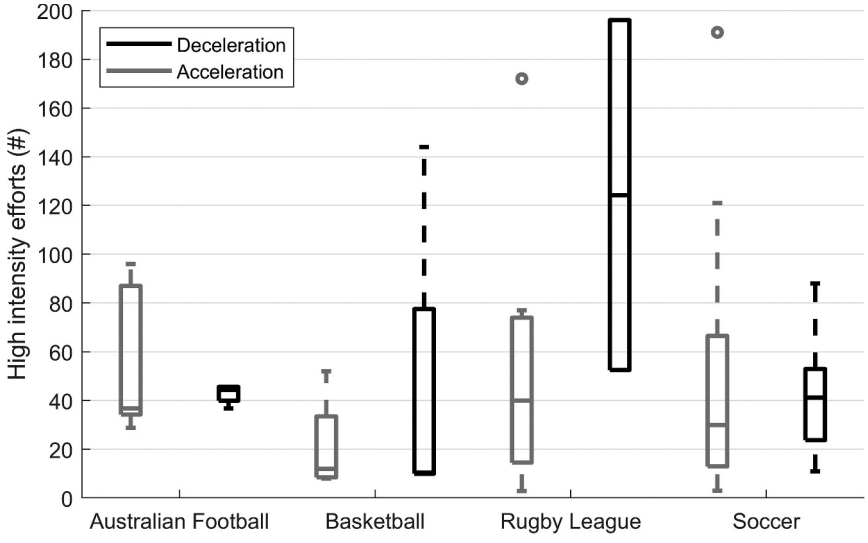

Figure 7. Range of reported averages of the number of high intensity acceleration and deceleration efforts (>2.78 m · s^{-2} or <−2.78 m · s^{-2}) per player per match (not position specific and only for players playing full time), categorised by sport.

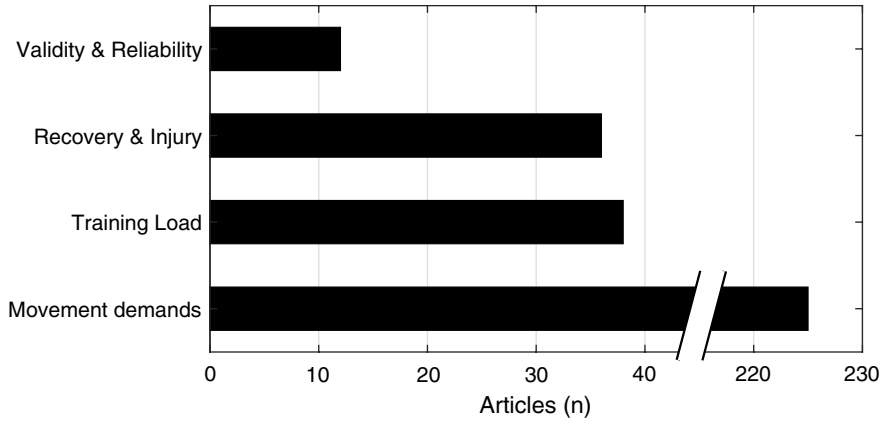

Figure 8. Application of acceleration and deceleration data and metrics of included articles.

and calibrated at a location before they can be used). However, it should be noted that GPS devices need a clear view of the sky to collect valid and reliable data; high walled stadiums can impede the view and therefore limit the use of GPS devices. The GPS devices also capture more information than an accelerometer does, thus making it a more commonly used tracking technology.

Details of manufacturer brand, model, sampling frequency and software version of athlete tracking technologies have not always been specified in research. It is hard to use and interpret data without the knowledge of the details of the athlete tracking technologies. Different manufacturer brands, models and software versions of athlete tracking technologies can have different sampling frequency, data processing procedures (deriving and cleaning) and technology inside, which directly change the quantified acceleration and deceleration metrics (Buchheit, Haddad et al., 2014; Varley et al., 2017). The sampling frequency of the tracking technologies determines the amount of data recorded per second. Researchers and practitioners should consider that if the sampling frequency is too low, higher frequency movements can be missed, however, too high could contaminate the signal with noise (Winter, 2009). Data that have been collected with accelerometers mostly sample at a higher frequency compared to GPS, LPS and optical tracking systems, at least 5 times higher in the studies included in this review, thus capture more data but are also more prone to noise. It has been reported that a 10 Hz GPS device is two to three times more accurate than a 5 Hz GPS device for measuring instantaneous velocity during acceleration and deceleration (M. Varley et al., 2012). However, GPS devices with a frequency above 10 Hz can be less accurate than 10 Hz devices (Johnston et al., 2014). This could be due to the fact that data processing will affect data output in addition to sampling frequency. Software versions of athlete tracking technologies can also highly influence the data, different software versions can have different data processing procedures and change the data output. Research has shown that maximal speed of optical tracking systems compared with that measured by a GPS, showed significantly higher maximal speed values. However, when the optical tracking system data was processed with the GPS software, no difference was reported (Ellens et al., 2021). This highlights the importance for researchers and practitioners to understand how their data is being processed and what factors (tracking technology manufacturer brand, model, sampling frequency and software version) may affect the final output. Further, the inclusion of this information in published research would allow comparisons of data from different projects.

The deriving techniques for acceleration and deceleration data were not specified in 66% of the included studies. It is important to know how the acceleration and deceleration data have been derived as different deriving techniques result in different data (Winter, 2009) and impact researchers' and practitioners' interpretation of the data. The derivation of acceleration and deceleration data from athlete tracking technologies can be done in three different ways. All athlete tracking technologies (except an accelerometer) can derive acceleration and deceleration data via a second derivative of position or first derivative of velocity. Modern GPS can also derive acceleration and deceleration data from Doppler-shift velocity (Malone et al., 2017). Doppler measurements are resistant to momentary signal anomalies and low signal-to-noise ratio caused by obstructions such as high concrete stands or roof of a stadium, making Doppler measurements a more accurate velocity measure than position-based velocity data (Chalko, 2007). Deriving the acceleration and deceleration data from either velocity (first derivative) or distance (second derivative) will result in different data. The first derivative increases the signals amplitude proportionally to the frequency, whereas the second derivative increases the signals amplitude proportional to the frequency squared (Winter, 2009) see, Figure 9. To put this into the context of tracking data, if displacement data contain frequencies of 10 Hz that have an amplitude of 1 m, the first derivative (velocity) will increase the amplitude to 10 $m \cdot s^{-1}$ (1*10). The second derivative (acceleration) will increase the amplitude to 100 $m \cdot s^{-2}$ ($1*10^2$), the amplitude of the acceleration signal is now one hundred times that of the displacement data. Such a rapid increase in amplitude has consequences for calculating acceleration via displacement (first derivative) or velocity (second derivative) when they have high-frequency noise present (more common when sampling with high frequencies). It is therefore important to consider the different deriving techniques of acceleration and deceleration data and the possible implications they may have on the data. Finally, acceleration and deceleration data can be accelerometer-derived. It is important to note that accelerometer-derived

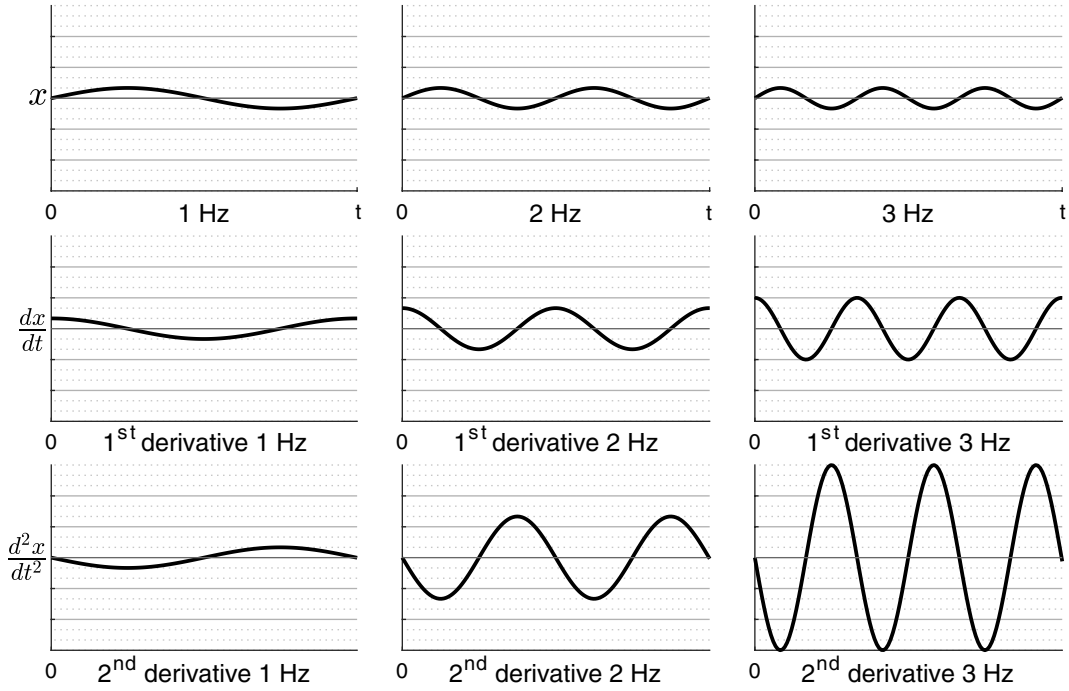

Figure 9. Changes in amplitude due to time differentiation of signals of increasing frequency. the first derivative increases the signals amplitude proportionally to the frequency, whereas the second derivative increases the signals amplitude proportional to the frequency squared. adapted from winter, d.a., biomechanics and motor control of human movement. 2009: John Wiley & Sons (p. 66).

data are different from ones derived from position or velocity data. An accelerometer measures acceleration in its own local frame in three directions (X, Y, Z) and is a direct measure of acceleration, whereas the other tracking technologies use a derived version of acceleration (derived from position or velocity data) which measures acceleration in a fixed frame. If it is not known which type of acceleration data is being used, incorrect conclusions may be drawn as the data is distinctly different. Therefore, the deriving techniques of the acceleration and deceleration data should always be mentioned as this will help the interpretation of data when making comparisons.

Most studies (92%) in this review did not mention using any cleaning techniques on the data. Of the studies using a cleaning technique, 48% used a low pass Butterworth filter. The low pass Butterworth filter passes signals with frequencies lower than a selected cut-off frequency and decreases signals (which are ideally noise) with frequencies higher than the cut-off frequency (Sinclair et al., 2013). The Butterworth filter requires a selected cut-off frequency, which will determine the amount of signal passed through the filter and noise blocked. This review found that the studies using a Butterworth filter used a range of cut-off frequencies between 0.02 and 15 Hz. The cut-off frequency can influence the data, but a rationale for selecting them is rare. The used rationale for selecting a cut-off frequency is visual inspection of residual analysis outputs and is based on a common technique used in biomechanics (Winter, 2009). The selection and influence of different filters and cut-off frequencies has been widely studied in the biomechanics literature (Day et al., 2021; Fazlali et al., 2020; Winter, 2009); however, limited research exists on the selection and influence of different filters and cut-off frequency on data obtained from athlete tracking technologies in team sports. Furthermore, as shown in this review, there is no rationale for using specific filters. As different data processing procedures (deriving and cleaning) on acceleration and deceleration data affect the resulting metrics (Winter, 2009), further research in this area is warranted.

Acceleration and deceleration data can be applied in various ways. The data can be used to monitor movement demands and training load, which are quantified using different metrics. Different criteria have been used within the literature to define metrics, such as acceleration thresholds and minimum durations, which the data need to meet to be included in the metrics. For example, criteria of a maximum acceleration effort defined at >3 $m \cdot s^{-2}$ for 0.5 seconds require the acceleration data to remain above 3 $m \cdot s^{-2}$ for a minimum of 0.5 seconds to be included as a maximum acceleration effort. It is important to consider data cleaning and set metric criteria when using efforts, as these can affect the number of efforts detected. The variation in reported high-intensity acceleration and deceleration efforts (>2.78 $m \cdot s^{-2}$ or < −2.78 $m \cdot s^{-2}$) across the literature may be explained by inconsistent data cleaning and set metric criteria. Strong data cleaning (e.g., filtering data over a large time interval) has been associated with less detected efforts for shorter minimum effort durations, compared to data cleaning with smaller time intervals (Varley et al., 2017). This difference disappeared when the end criteria of an effort, specified by the acceleration reaching zero, were included. When an effort is defined by just a threshold or a threshold and minimum duration, more efforts can be detected than have been performed. For example, when the data meet effort criteria defined at >3 $m \cdot s^{-2}$, then fall below the threshold for just one sample and rise again above the threshold followed by researching zero, two efforts will be detected. However, just one effort has been performed. To account for the fluctuation of the acceleration

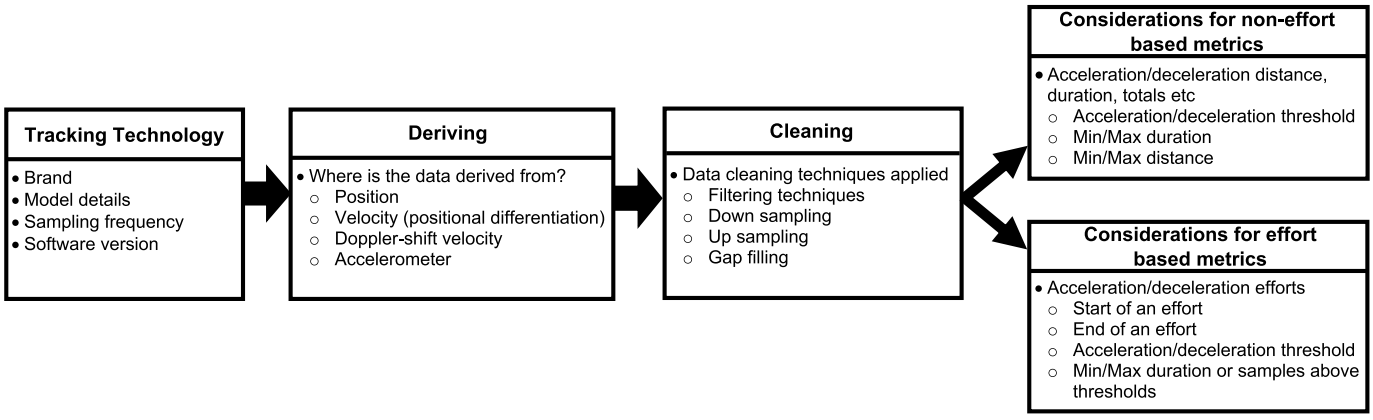

Figure 10. Flow chart of the considerations that will affect acceleration and deceleration data and metrics. ● = Information that should be mentioned when using acceleration and deceleration data, ○ = Examples of the information that should be provided. The word *data* in the figure is referring to acceleration and deceleration data and the word *metric* is referring to acceleration and deceleration metrics.

data round the set threshold, an end criterion for the efforts is needed. It has been shown in this scoping review that majority of the studies using number of efforts performed, only specified the effort by a set threshold. Caution should be taken when comparing metrics involving set criteria. Studies with similar effort threshold names can have different threshold criteria, for example, a medium intensity acceleration effort can be classified with thresholds between 1.5 and 3 $m \cdot s^{-2}$. For more in-depth analysis of threshold-based metrics, the inclusion of initial velocity can be utilised. As the magnitude of acceleration decreases with increasing initial velocity (Sonderegger et al., 2016), it is rarer for an acceleration of a given magnitude to occur at higher velocities. Initial velocity could provide extra context for how a player achieved a given effort, for example, was the effort achieved from a running start or standing still. This information might be relevant for research needing extra context to the effort metric. Future research should explore initial velocity further and include transparent criteria to define acceleration and deceleration metrics, especially when using efforts. In summary, it is important to consider the criteria used to define acceleration and deceleration metrics, and future research investigating new criteria such as the use of set distances covered and steps taken are required.

The data can also be used to investigate the validity and/or reliability of data quantified metrics. When using the data for validity and/or reliability purposes, it is important to know the effect of data cleaning and set criteria on the used metrics, as this will directly influence the data and thus the validity and/or reliability of the metrics. The validity and reliability of acceleration and deceleration metrics have been shown to be prone to error, especially at higher velocities (Crang et al., 2021; M. Varley et al., 2012), and can be substantially affected by data cleaning (Delaney, 2019). There is currently no research investigating which data cleaning method provides the most valid and reliable acceleration and deceleration output. Therefore, careful consideration to the data cleaning used is needed, and future research should investigate which data cleaning methods provide the most valid and reliable acceleration and deceleration output. Practitioners and researchers should be aware of the differences between studies in the criteria used to define metrics when comparing findings, and the effect the criteria and data cleaning could have on their results.

This scoping review is the first to systematically map the literature around the techniques and rationale used to derive and clean acceleration and deceleration data obtained from athlete tracking technologies in team sports. Lastly, all considerations that could affect acceleration and deceleration data (that have been mentioned in this scoping review) have been summarised in Figure 10. When using acceleration and/or deceleration data, practitioners and researchers should consider all these factors, as different choices will affect their data differently and may result in different outcomes. This scoping review has some limitations that should be recognised and considered when interpreting the findings of this review. The review specifically focussed on team sports; therefore, application in other sports has not been explored. Furthermore, this review did not examine the quality of the articles included and was limited to articles published in English.

5 Conclusion

Using a systematic approach, this scoping review identified articles that have explored the use of acceleration and deceleration data obtained from athlete tracking technologies in team sports and mapped the techniques and rationales used to derive and clean data. The scoping review identified research gaps in the methodological procedures, as well as deriving and cleaning techniques that warrant future research focussing on its effect on acceleration and deceleration data. A research gap was identified in the data processing procedures, specifically, on how acceleration and deceleration data is derived from athlete tracking technologies and what sequence of steps have been taken to clean the data. Further, it has been identified that method sections are lacking specific information regarding the used athlete tracking technologies, which prevents replicability of the studies. It is

therefore recommended that studies using acceleration and deceleration data obtained from athlete tracking technologies in team sports report the athlete tracking technology manufacturer brand, model details, sampling frequency used and software version. Furthermore, the used deriving and cleaning technique of the acceleration and deceleration data should be reported together with the used criteria to define metrics. The inclusion of this information in published research would allow comparisons of data from different projects so meaningful conclusions can be drawn.

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ORCID

Susanne Ellens http://orcid.org/0000-0003-1336-8560
Kane Middleton http://orcid.org/0000-0002-4914-8570
Paul B. Gastin http://orcid.org/0000-0003-2320-7875
Matthew C. Varley http://orcid.org/0000-0003-3033-571X

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